

IMEN JARRAYA

Doctor in Electrical Engineering | Research Developer | University Teacher

Email imenjarraya85@gmail.com

Location Riyadh, Saudi Arabia

Email ijarraya@psu.edu.sa

Affiliation Prince Sultan University (RIOTU)

Phone +966 55 725 44 96

Nationality Tunisian

PERSONAL STATEMENT

Doctor in Electrical Engineering and Postdoctoral Fellow with strong expertise in hybrid electric vehicle energy management, embedded systems, autonomous UAV navigation, computer science, and IoT/AI-based intelligent systems. She combines research and development experience with skills in programming, computational modeling, hardware integration, and intelligent system design. She also has extensive experience in university teaching, laboratory supervision, student mentoring, and final-year engineering project supervision.

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow — Robotics and Internet of Things Unit (RIOTU)

Oct 2023 - Present

College of Computer & Information Sciences (CCIS), Prince Sultan University · Robotics Group · Riyadh, Saudi Arabia

- ▶ Conduct advanced research in embedded systems, UAVs, robotics, artificial intelligence, and IoT applications within RIOTU.
- ▶ Develop UAV localization and navigation frameworks for GNSS-denied environments using sensor fusion, terrain-aided navigation, deep learning, ROS 2, PX4, and Gazebo.
- ▶ Design AI-based models for autonomous systems, including CNN-LSTM-Attention, KAN-based architectures, EKF-based fusion, and trajectory-prediction methods.
- ▶ Contribute to UAV dataset generation, simulation environments, and experimental validation for autonomous navigation and aerial tracking.
- ▶ Participate in research projects on UAV navigation, drone-to-drone tracking, quadruped robotics, embedded AI, and intelligent decision-making.
- ▶ Publish high-impact research articles and support student mentoring, research collaboration, and scientific activities within RIOTU.

University Teaching Assistant

2021 - 2023

Higher Institute of Technological Studies (ISET), Sidi Bouzid, Tunisia

- ▶ Delivered lectures and laboratory sessions in Real-Time Systems, Embedded Systems Programming, and Internet of Things.
- ▶ Designed course materials and supervised end-of-studies graduation projects.
- ▶ Co-supervised a Ph.D. thesis on electric vehicle energy optimization; served on multiple assessment committees.

Research and Development Engineer

2020 - 2021

Novel-Ti · Sfax, Tunisia

- ▶ Led R&D in embedded systems, robotics, IoT, and applied artificial intelligence.
- ▶ Designed PCBs for a gateway module and thermistor-based food sensor using Altium Designer.
- ▶ Developed a children's gaming platform in Python.
- ▶ Implemented hoverboard tilt control using an MPU6050 gyroscope and GD32F13 microcontroller.

University Teaching Assistant

2011 - 2015

Higher Institute of Industrial Management of Sfax (ISGIS), Tunisia

- ▶ Taught Industrial Computing Workshop, Electronic & Electrotechnical Workshop, and Basics of Automation.
- ▶ Supervised and co-supervised undergraduate projects in electronics, photovoltaics, and automation.
- ▶ Contributed to the development of course curricula and laboratory exercises.

University Teaching Assistant

2010 - 2011

Higher Institute of Biotechnology - Sfax, Tunisia

- ▶ Delivered the Electrical Engineering Workshop and Biomedical Measurement course (Master level, KL-72001 platform).

EDUCATION

Ph.D. in Electrical Engineering — Mention: Very Honorable

9 Jan 2021

National School of Engineers of Sfax (ENIS) & ESTACA, France

- ▶ Thesis: Energy management of an electric vehicle fitted with an on-board hybrid energy storage unit (Li-ion battery + supercapacitors).
- ▶ Laboratory: CES_Lab — ENIS & S2ET-ESTACA. Supervisor: Prof. Hamed Trabelsi.
- ▶ Co-supervisors: Prof. Nassim Rizoug & Mr. Mohamed Hedi Chabchoub (ESTACA).

Research Master's — Electrical Conversion & Renewable Energies

2008 - 2010

National School of Engineers of Sfax (ENIS)

- ▶ Thesis: Development of a decision support tool for wind power systems adapted to Tunisian sites (July 2010).

National Diploma in Electrical Engineering

2006 - 2009

National School of Engineers of Sfax (ENIS)

- ▶ Final project: Contribution to the modeling of a wind turbine chain connected to the single-phase network.
- ▶ Final rank: 13.92 / 20.

Diploma of First Cycle University Studies — Physics / Chemistry

2004 - 2006

Preparatory Institute for Engineering Studies, El Manar

Scientific Baccalaureate — Experimental Sciences

2000 - 2004

Mahmoud Magdich High School, Sfax

- ▶ Final rank: 15.45 / 20.

SELECTED RESEARCH PROJECTS

QuadroRL — Robust Terrain Traversal for Search & Rescue Quadrupeds

2026

RIOTU Lab, Prince Sultan University - Riyadh, Saudi Arabia

- ▶ Developing reinforcement-learning-based locomotion policies for quadruped robots in unstructured SAR terrains.
- ▶ Integrating simulation-to-real transfer strategies using ROS 2 and physics-based simulation environments.

Agile Target Tracking — Drone-to-Drone Tracking Framework

2022 - 2023

Prince Sultan University · Riyadh, Saudi Arabia

- ▶ Contributed to a research project on real-time tracking of agile aerial targets, with a focus on Unmanned Aerial Vehicles (UAVs).
- ▶ Co-developed D2DTracker — a real-time trajectory-prediction framework for drone-to-drone tracking (UVS 2023, Oman).

PSDSARC Project — Terrain-Aided Navigation (TAN)

2023 - 2025

Prince Sultan University · Riyadh, Saudi Arabia

- ▶ Designed TAN algorithms and a Two-Stage Residual EKF (XGBoost + KAN) for autonomous UAV localization in GPS / GNSS-denied environments.
- ▶ CLAK (CNN-LSTM-Attention + KAN): Developed a novel architecture for non-visual UAV localization — published in *Satellite Navigation* (Q1, IF 10.1), 2026.

iHunter Project — Unified UAV Aerial Tracking Framework

2023 - 2025

Prince Sultan University · Riyadh, Saudi Arabia

- ▶ Built a unified UAV aerial-tracking framework integrating SMART-TRACK (Kalman filtering + visual detections) with a GRU predictor trained on FLIGHTGEN.
- ▶ Validated end-to-end in ROS 2 / PX4 / Gazebo simulation environments for robust drone-to-drone target tracking.

Power Management Strategy for Hybrid Electric Vehicles

2022 - 2023

King Abdulaziz University · Jeddah, Saudi Arabia

- ▶ Conducted experimental studies to optimize energy management strategies for hybrid electric vehicles.
- ▶ Collaborated with KAU on advancing power management architectures for HEV systems.

Smart Mailbox — IoT & AI Automation

2020 - 2021

Smart Ways Innovations · Sfax, Tunisia

- ▶ Developed an IoT/AI-based mail management system featuring real-time sensors and intelligent sorting.
- ▶ Designed a public-lighting control and monitoring solution with IoT/AI-based fault management.

PUBLICATIONS — 15 JOURNAL ARTICLES • 11 CONFERENCE PAPERS

Journal Articles

- [J1] Tayari K., Jarraya I., et al.. WTIS: Wearable Thoracic Surface Impedance-Based Sensor for Continuous Cardiac Monitoring. *SSRN* 6269315 • 2026
- [J2] Jarraya I., Abdelkader M., Gabr K., et al.. CLAK: CNN-LSTM-Attention and KAN for Non-Visual UAV Localization in GNSS-Denied Environments. *Satellite Navigation* (Q1, IF 10.1) • 2026
- [J3] Abdelkader M., Jarraya I., Gabr K., et al.. A Two-Stage Residual EKF using XGBoost and KAN for Terrain-Aided UAV Localization in GNSS-Denied Environments. *Engineering Applications of AI* (Q1, IF 8.0) • 2026 • vol. 173, p. 114430
- [J4] Jarraya I., Gabr K., Abdelkader M., et al.. A Comparative Study of TERCOM-EKF and ViSensorRF for UAV Localization in GNSS-Denied Environments. *Results in Engineering* (Q1, IF 7.9) • 2025 • p. 108279
- [J5] Jarraya I., Abdelkader M., Gabr K., et al.. OS-RFODG: Open-Source ROS 2 Framework for Outdoor UAV Dataset Generation. *Results in Engineering* (Q1, IF 7.9) • 2025 • p. 106613
- [J6] Jarraya I., Atitallah S.B., Alahmed F., et al.. SOH-KLSTM: A Hybrid Kolmogorov–Arnold Network and LSTM for Enhanced Li-Ion Battery Health Monitoring. *Journal of Energy Storage* (Q1, IF 9.8) • 2025 • vol. 122, p. 116541

- [J7] Jarraya I., Al-Batati A., Kadri M.B., et al.. GNSS-Denied UAV Navigation: Analyzing Computational Complexity, Sensor Fusion, and Localization Methodologies. **Satellite Navigation (Q1, IF 10.1) • 2025 • vol. 6, no. 1, p. 9**
- [J8] Abdelhedi F., Jarraya I., et al.. Optimizing EV Efficiency with Hybrid Energy Storage: Comparative Analysis of Rule-Based and Neural Network Power Management. **Energy (Q1, IF 9.4) • 2024 • vol. 313, p. 133979**
- [J9] Tayari K., Jarraya I., Cheikhrouhou O., et al.. Surface Impedance for Real-Time Cardiac Monitoring in Body Area Networks. **IEEE Access (Q1, IF 3.6) • 2024 • vol. 13, pp. 93222–93234**
- [J10] Ben Youssef M., Jarraya I., Zdiri M.A., Ben Salem F.. Support Vector Regression-Based SOC Estimation for Batteries: Cloud vs Non-Cloud. **Indonesian Journal of Electrical Engineering & Computer Science • 2024 • vol. 34, no. 2, pp. 697–710**
- [J11] Abdelkader M., Gabr K., Jarraya I., et al.. SMART-TRACK: A Novel Kalman Filter-Guided Sensor Fusion for Robust UAV Object Tracking in Dynamic Environments. **IEEE Sensors Journal (Q1, IF 4.5) • 2024 • vol. 25, no. 2, pp. 3086–3097**
- [J12] Jarraya I., et al.. Comparison Study Between Hybrid NM-PSO and OCV-RLS for Battery Parameters Estimation. **Journal of Energy Storage (Q1, IF 9.8) • 2022 • vol. 50, p. 104424**
- [J13] Jarraya I., et al.. An Online State of Charge Estimation for Li-Ion and Supercapacitor in Hybrid Electric Drive Vehicle. **Journal of Energy Storage (Q1, IF 9.8) • 2019 • vol. 26, p. 100946**
- [J14] Jarraya I., Ammous K.. Development of Decision Support Tool for Wind Energy Systems Adapted to Tunisian Sites. **Int. Journal of Advancements in Research & Technology • 2016 • vol. 5, issue 6**

Conference Papers

- [C1] Alwajih F., et al. (incl. Jarraya I.). PALM: A Culturally Inclusive and Linguistically Diverse Dataset for Arabic LLMs. **ACL 2025 • Long Papers — Best Resource Paper • 2025**
- [C2] Abdelkader M., Al Batati A., Jarraya I., Koubaa A.. D2DTracker: Real-Time Trajectory Prediction for Agile Drone-to-Drone Tracking. **2nd Int. Conf. on Unmanned Vehicle Systems (UVS), Oman • 2023**
- [C3] Jarraya I., Abdelkader M., Gabr K., et al.. Non-Visual Terrain Navigation for UAVs: A Comparative Study in GPS-Denied Environment. **SMARTTECH 2024, Marrakesh, Morocco • 2024**
- [C4] Abdelkader M., Gabr K., Nacar O., Jarraya I., Koubaa A.. FLIGHTGEN: ROS 2-Powered Automated UAV Dataset Generator. **SMARTTECH 2024, Marrakesh, Morocco • 2024**
- [C5] Hmad J., Jarraya I., et al.. An Improved Control Strategy of Stand-Alone PV, Wind, and Battery System. **20th SSD Conference, IEEE • 2023**
- [C6] Hmad J., Jarraya I., et al.. Power Management Strategies of Hybrid Storage System Supplying Electric Vehicle. **30th Mediterranean Conf. on Control & Automation (MED), IEEE • 2022**
- [C7] Jarraya I., et al.. Energy Management Strategy Based on Fuzzy Logic for EV with Hybrid Source. **19th SSD Conference, IEEE • 2022 • pp. 347–354**
- [C8] Jarraya I., et al.. An Online Grid Impedance Estimation Using Recursive Least Square for Islanding Detection. **16th SSD Conference, IEEE • 2019 • pp. 193–200**
- [C9] Jarraya I., et al.. Modeling and Parameters Estimation for Li-Ion Cells in Electric Drive Vehicle. **15th SSD Conference, IEEE • 2018 • pp. 1128–1132**
- [C10] Jarraya I., et al.. Modeling and Simulation of an On-Grid Variable Speed Aerogenerator. **7th Int. Renewable Energy Congress (IREC), IEEE • 2016**
- [C11] Jarraya I., et al.. Choix des systèmes éoliens appropriés pour les sites tunisiens. **IREC • 2010 • pp. 5–7**

Books & Software

- **Book proposal (in preparation):** Embodied Agents — The New Convergence of AI and Robotics.

- **Software:** *gps_denied_navigation_sim* — Open-source PX4 SITL + Gazebo benchmark for GPS-denied UAV navigation.

Research in the Media

The Satellite Navigation paper [J7] on GNSS-denied UAV navigation ranked in the 96th percentile of articles of similar age across all journals on Altmetric, and was featured by international science and technology platforms:

- TechXplore — “Smart skies: New methods for UAVs to navigate in GPS-denied environments” (April 2025).
- Electronics For You — “UAV Navigation in GPS-Denied Environments.”
- AZoSensors · Bioengineer.org · DroneStacks (LinkedIn) — Dedicated coverage of the publication's findings on UAV navigation strategies.

TECHNICAL SKILLS

Domain	Tools & expertise
Simulation	MATLAB / Simulink / Stateflow, LabVIEW, dSPACE, ControlDesk
Programming	C / C++, Python, Visual Basic, Android
Robotics / ROS	ROS 1 & ROS 2, Gazebo, PX4 Autopilot
Hardware	Arduino, Raspberry Pi, STM32F4, ESP32, Jetson Nano
EDA / CAD	Altium Designer, AutoCAD
Industrial PLCs	Grafcet programming
Build / Environment	Makefile, CMake, CLion, Linux / Ubuntu, Windows
Office / Documentation	LaTeX, MS Office Suite
Languages	Arabic (Native) · French (Fluent) · English (Professional)

UNIVERSITY SUPERVISION & EXAMINED PROJECTS

Doctoral Co-Supervision

- Ph.D. thesis co-supervision (ENIS, Sfax, since 2019) — Contribution to the optimization of electric vehicle routing for minimal electrical energy consumption.

Graduation Projects Supervised

- **2021 – 2022 (ISET Sidi Bouzid):** Intelligent IoT/AI-based airport intrusion monitoring system · Design and realization of a Connected Mailbox · Surveillance & Security Drone.
- **2015 – 2016 (ISGIS Sfax):** Arduino acquisition card for measuring and transmitting data from photovoltaic panels (Applied Bachelor, Electronics & Automation).
- **2015 – 2016 (ISGIS Sfax, co-supervision):** Electronic card for control of physical quantities of a photovoltaic installation.
- **2014 – 2015 (ISGIS Sfax, co-supervision):** Study and design of a charge regulator for an autonomous photovoltaic system.

Graduation Projects Examined (Rapporteur / Jury Member)

- ▶ **2021 – 2022:** Detailed study of the electrification of a group of subscribers in Sidi-Bouزيد Ouest · Study and construction of an electrical installation for a new factory.
- ▶ **2015 – 2016:** Central gas and fire detection system · Design of a SEGWAY robot (Electronics, Electrotechnics & Automation).
- ▶ **2012 – 2014:** Automatic counting system of vaccine vials and serums in the filling chain.
- ▶ **2011 – 2013:** Access system controlled by PIC microcontroller · Lead battery charger · Electronic card for electrical signal acquisition · Adaptation of priming pump on shunting locomotives.

INTERNSHIPS

Thesis Internship

2019

ESTACA 2 · France

- ▶ HIL-based embedded energy management for a hybrid EV (Li-ion + supercapacitor); PSO-NM and RLS battery model identification; validated on a MATLAB / dSPACE test bench.

Engineer Internship

Sept 2016

SITEM Somfy Group · Hammam Zriba, Tunisia

- ▶ Motor manufacturing workshops; engine diagnosis, disassembly, repair, reassembly, and quality control. Administrative management of parts and laboratory.

Technician Internship

July 2008

Universal Engineering Company · Tunisia

- ▶ Specialized studies in electricity safety: fire protection and environmental safety.

Worker Internship

July 2007

Universal Engineering Company · Tunisia

- ▶ Wiring of electrical cabinets; specialized automation of electrical panels.

CERTIFICATIONS & RECENT TRAINING

- ▶ **Nov 2025** — Compliance Tools to Success Workshop, Compliance and Policies Office, Prince Sultan University.
- ▶ **Oct 2025** — Contributor to the 9th Students Research Forum & Research Labs Expo 2025, Prince Sultan University.
- ▶ **Apr 2025** — Science Research Forum IV (Department of Mathematics & Sciences, PSU) · Policy Development Process Workshop (Compliance, Legal & Policies Office) · Seminars on Few-Shot/Self-Supervised Learning and iHunter UAV Framework (RIOTU Lab).
- ▶ **Mar 2025** — LEAP 2025 International Technology Conference, Riyadh, Saudi Arabia.
- ▶ **Feb 2025** — 8th CDMA International Conference on Data Science and Machine Learning Applications.